

Day 8 - Data Quality & Class Audit



Data Quality Table

Metric	Value
Total Rows	399
Total Columns	6
Missing Values	0
Duplicate Rows	0
Invalid Bit Labels	0
Class 0 Samples	242
Class 1 Samples	157
Class 0 Percentage	60.65%
Class 1 Percentage	39.35%
Class Balance Status	Moderately Imbalanced

Obseravtion :

- Dataset contains 399 samples and 6 features.
- No missing values were found.
- No duplicate records were detected.
- No invalid bit labels were present.
- Target classes are moderately imbalanced (Class 0: 60.65%, Class 1: 39.35%).
- Overall, the dataset is clean and ready for preprocessing and model training.

Cleaning Plan :

- 1.Drop the S No column as it does not contribute to prediction.
- 2.Verify and standardize NTU labels if multiple formats exist.
- 3.Ensure Analog Value and Analog Voltage are stored as numeric data types.
- 4.Check for class imbalance and use stratified train-validation-test splitting.
- 5.Preserve the distribution of both Binary bit S classes and NTU levels across all splits.
- 6.Scale numerical features if required by models such as Logistic Regression or SVM.

